

Where both papers stand — status before the 23 September meeting

Cowork note for Chris — HardDisks / hspist3

2026-09-23 (rev. 3, τ_T measured)

The two sentences

Paper 1 — speed of sound

We measure the adiabatic sound speed of a 2D hard-disk fluid from the resonance of a heavy divider, over $\eta = 0.0065\text{--}0.76$ with nine divider masses and 25 seeds per point. At $N = 100$ the curve sits $+1.0\%$ above Kolafa–Rottner on average, the offset shrinks with N , and we show that Román et al. (2002) — same box, same N , 24 years ago — have exactly the same $+1\%$ once their data are mapped with the correct adiabatic formula instead of their $\gamma = 2$.

Paper 2 — energy transfer

A proof ladder for where injected work goes in a hard-disk box: Levels 0–2 closed (ledgers, quasi-static work, the u^2 excess and its friction), Level 3 closed on the observable (a pre-loaded spring wall settles where a parameter-free model says, to -1.3% with the box’s own measured pressure, $+3.6\%$ with the bulk EOS), Level 4 measured in both halves (a free divider passes exactly half a slow push’s work at any mass; the thermal relaxation through it is the adiabatic-piston problem, and its measured timescale selects the hard-disk theory at -0.6σ and excludes the ideal gas at 3.5σ). The levels are tied to Paper 1 by one line: the gas spring is $k_{\text{gas}} = Nmc_s^2/L^2$.

1 Paper 1 — what is established

Item	Status and number
Apparatus	EDMD, $k_B T = m = \sigma = 1$, $N_s = 50$ per compartment, $h = 10$, divider thickness $t = 0.05$; event ledger to 10^{-12} . [DATA]
Estimator	Periodogram of the divider position, largest bin above $\nu_{\text{pred}}/2.5$; bin floor 0.144%; flat for any floor in $[\nu_{\text{pred}}/6.25, \nu_{\text{pred}}/2.08]$; largest-bin bias across the mass ladder $-0.030 \pm 0.029\%$ per e-fold (flat); the resonance-fit alternative leans -0.29% per e-fold (14σ) and was rejected on that evidence. [DATA]
Theory	$\cot K = MK/(2Nm)$, $\nu_1 = c_s K/(2\pi L_{\text{eff}})$, $c_s^2 = (k_B T/m)(Z + \eta Z' + Z^2)$ — all three derived from first principles in the 20 Sept note; c_s is the slope of ν_1 vs $K/2\pi L_{\text{eff}}$ through the origin over nine masses. [DERIVATION]
Effective length	$L_{\text{eff}} = L_0 - 2r - t/2$; the missing $t/2$ was a real error (-0.004% at low η , -0.50% at $\eta = 0.65$), now a single constant with an assert guard. [DATA]
Canonical result	260919 set: $+1.04\%$ above KR averaged over $\eta \leq 0.4$; $+2.6\%$ at $\eta = 0.52$; points beyond $\eta \approx 0.69$ are in the ordering region and are shaded, not counted. [DATA]
Finite size	Offset shrinks with N (A2 ladder); Z_{wall} as an explanation withdrawn (overpredicts $5\times$); $1/\sqrt{N}$ extrapolation as a surface-scaling hypothesis. [DATA]
Román 2002	His Table I re-mapped adiabatically against KR: $+1.0, +0.9, +1.5, +0.9, +1.3, +1.1, +1.0\%$ — flat, same as ours; his own $\gamma = 2$ mapping made it look like $+1.3 \rightarrow +10.6\%$ growing with density. His SPT/Henderson columns reproduce to every printed digit from our EOS code, which validates both. His N -series $3.81 \rightarrow 3.74$ falls the same way as ours. [DATA]
New this week	The box's pressure, measured directly on a held wall (Paper 2's $F(L)$ set), is $Z_{\text{box}}/Z_{\text{KR}} = 1.0248 \pm 0.0014$, flat over $\eta = 0.100$ – 0.111 . Through the bulk formula that would imply $+1.9\%$ in c_s ; the divider mode measures $+1.0 \pm 0.3\%$. So the 100-disk box is 2.5% over-pressured and 1% over-stiff, and the two are not the same number — a finite-size fact worth one paragraph. [OPEN]
Draft	IEEEtran skeleton, 3 pages, Román section and table in. Full v1 (6–8 pages, every number sourced to a CSV) is tonight's CC job; no runs needed.
Left for KOA	$N \approx 1000$ sweep and the $\eta = 0.65$ size series. Account expected Friday 25 Sept.

Paper 1

Submit with the $N = 100$ curve, the finite-size ladder and the Román re-mapping as the story, and add the KOA $N \approx 1000$ points as a revision — or hold until KOA? My recommendation: write v1 now, decide after the first KOA week; the draft does not change shape either way. Second question: which venue (the Román paper is Am. J. Phys.; the result is Phys. Rev. E / J. Chem. Phys. material).

2 Paper 2 — the ladder, level by level

Level	Status	What it says, with numbers
0	closed	Event ledgers close to $10^{-12}/10^{-13}$; the master box with both walls held reproduces geometry A to 0.11σ (Level 0b), so Levels 0–2 are valid in the master box.
1	closed, pass 0.3σ	Quasi-static work for a 10 % compression ($\Delta x = 3.93\sigma$, η 0.100 $\rightarrow$ 0.111): $W_{\text{qs}}^{\text{finite}} = 7.4685 \pm 0.0131 k_{\text{B}}T$; $T_f/T_i = 1.149$. Independently reproduced this week by the $F(L)$ set in a different geometry: $T = 1.1499$ at 10 %.
2	closed, honest verdict	“Passes on mean work at both ends; prediction corrected mid-level; linear friction consistent with zero at $0.32u$; variance 25–40 % below Poisson unexplained.” $A_{\text{step}} = 25.9 \pm 4.3$, $A_{\text{ramp}} = 11.3 \pm 2.5$; friction from Paper 1’s linewidth $\Gamma = 0.31\text{--}0.33$ vs Enskog 0.331; fast end $\langle W \rangle = 2N_s m(\Delta x/L)u^2$ to 0.986.
3	closed on the observable	See §2.1. Settled wall displacement $\bar{s} = 0.8705 \pm 0.0049\sigma$; model with bulk KR 0.8404 (+3.6 %, 6σ); model with the box’s measured $F(L)$ 0.8819 (–1.3 %, 2.3σ). Attribution of the last per cent open.
4	two halves, one measured	<i>Mechanical half (measured)</i> : a free divider between two 50-disk gases passes exactly half a slow push’s work and takes half the compression at every mass over $M_d = 10\text{--}200$ (far/total 0.49 ± 0.13 , 0.44 ± 0.13 , 0.28 ± 0.15 vs $1/2$; settled $1.87\text{--}2.25\sigma$ vs $\Delta x/2 = 1.965$). Because it follows quasi-statically it creates no temperature difference, so thermal transmission is a separate experiment. <i>Thermal half (the adiabatic-piston problem, Lieb 1998)</i> : theory from Gruber–Piasecki 1999 Eq. 51 and Cencini et al. 2007 gives $\tau_T = (4/\sqrt{2\pi}) ML_c/\sqrt{mkT}$, linear in M ; carrying the hard-disk EOS through the isobar shortens it by 1.228 to $\tau_T = 50M$ (ideal gas $62M$). A temperature-step run (20 seeds) shows stage 1 (the divider swings to the KR-predicted pressure-balance offset 3.99σ and rings) but cannot resolve stage 2: the per-seed spread of $T_1 - T_2$ is the microcanonical energy-split fluctuation, 0.198 measured vs 0.199 predicted, i.e. the physics, not noise. So τ_T was measured instead from <i>equilibrium</i> fluctuations (fluctuation–dissipation, no step), with the analysis pre-registered and its bias calibrated on synthetic series before any real record was fitted. Result at $M_d = 10$, 80 seeds, 65 τ_T of record: $\tau_T = 491 \pm 29$ from $T_1 - T_2$ and 487 ± 29 from the divider position, against 503.5 (hard-disk isobar) and 618 (ideal gas) — the hard-disk prediction at -0.6σ, the ideal gas excluded at 3.5σ . The oscillatory part of the same autocorrelation has period $93\text{--}96 \pm 5$; Paper 1’s $\cot K = \alpha K$ at $\alpha = 0.1$ predicts 95.2 with KR (ideal gas 117, excluded at 5σ on its own). The mode’s damping, $\tau_r = 207 \pm 4$, is $2\times$ faster than the Mansour piston form (426) — the one Level 4 number that disagrees with an independent prediction. The exponent in M is untested at one mass; the four-mass ladder ($M_d = 20\text{--}200$) runs on the Mac the night before the meeting (the $M_d = 10$ cell took 13 minutes, not the night I had budgeted). Two papers, one autocorrelation function.
5–6	next	The stationary heat-conduction design (per-wall heat baths, a gated ledger change, tests Eq. 52 directly); chain of dividers. KOA: Paper 1’s $N \approx 1000$ sweep and the $\eta = 0.65$ size series.

2.1 Level 3 in five lines, because it will be asked

The apparatus is a pre-loaded spring: it already holds the standing gas force F , so the energy it captures is $\Delta E = Fs + \frac{1}{2}ks^2$, first order in the wall displacement s — weight-lifting work, with $k \rightarrow 0$ a pushrod and $k \rightarrow \infty$ a rigid wall. [\[DERIVATION\]](#) The gas is an adiabatic spring of stiffness $k_{\text{gas}} = Nmc_s^2/L^2 = 0.0494$, which is also the heavy-divider limit of Paper 1’s resonance equation — the two papers share this line. [\[DERIVATION\]](#) The model is one degree of freedom with no fitted parameter, $M\ddot{s} = F_{\text{ad}}(L - x_p + s) - F_{\text{ad}}(L) - ks$. It predicts where the wall *settles*: measured

$0.8705 \pm 0.0049 \sigma$, model 0.840 with bulk KR, 0.882 with the box’s own measured pressure. **[DATA]** The “transient excess” of 19–57 % reported in v4 was a peak-picking statistic: the identical statistic on a no-push control gives the same number (+0.463 vs +0.446 σ), and it does not scale with u as an acoustic pulse must. Withdrawn. **[DATA]** What is open is the last per cent: with the wall’s actual pressure the model is 2.3σ low. My reading: the wall force is the box’s real pressure (wall theorem, $P = k_B T \rho_{\text{contact}}$), so the $F(L)$ route is the physically right input for a wall, and the remaining -1.3% is a genuine small residual, not a bracket between two wrong inputs. **[OPEN]**

Paper 2

Is “Level 3 closes on the observable: the model predicts the settled displacement to $-1.3 \pm 0.6\%$ with the box’s measured pressure” an acceptable close, or do we chase the last per cent before Level 4? My recommendation: accept, state it, move to Level 4 on KOA; the per cent is a finite-size effect of the same family as Paper 1’s and belongs in one shared paragraph. Second: the Level 4 ladder — $M_d \in \{10, 50, 200, 1000, 5000\}$, 40 seeds, records $5\tau_{\text{heat}}$ — is the first KOA campaign for Paper 2. OK?

Paper 2 Level 4 — scope

Level 4 has turned into a test of the adiabatic-piston (Lieb) problem in an *interacting* gas, which Cencini et al. call “still too ambitious” with “very few studies”. Is that now the centre of Paper 2, with Levels 0–3 as the validated apparatus, or does it stay one section of the “where does injected work go” ladder? KOA is now Paper 1’s N -sweep only: the Level 4 mass ladder turned out to be a Mac night. Second: building per-wall heat baths for the stationary-conduction experiment is a change to the energy ledger (gated, Level 0 re-run) — worth it?

3 Errors caught and fixed — in case “how do you know” comes up

Each was found by a check we now run every time: the missing $t/2$ in L_{eff} (assert guard); the isothermal gas spring replaced by the adiabatic one (tied to c_s); a normalisation bug in the Level 3 adiabat that made the drive 28 % too weak (fixed-point check against a standalone calculation); the peak statistic (no-push control); the Z_{wall} explanation (measured, overpredicts $5\times$); the Level 4 KE columns that silently summed both gases in the master box (two-compartment box, no code change). Every claim in the plan carries a tag: SOURCE, DERIVATION, INFERENCE, DATA or OPEN.

4 Next two weeks

- Done since Monday: Paper 1 draft v1 (4 pp), Paper 2 draft v1 (4 pp), figure pack, Level 3 closed, Level 4 both halves, adiabatic-piston papers read and cited.
- Thu 24: meeting. Decisions above.
- Fri 25: KOA account. First hour: ssh, hello job on sandbox, rsync via DTN to `koa_scratch`, build with `-march=x86-64-v2`, one cell, a 10-task array on `shared,kill-shared,seff`.
- Week of 28 Sept: Paper 1 KOA campaign ($N \approx 1000$ sweep, $\eta = 0.65$ size series); Level 4 ladder. Paper 1 v1 to Susanne by Monday 28.
- Repo hygiene, not before deadlines: `git filter-repo` for the 25 GB szilard blobs; `git gc` for `tmp_pack_*`.